

MINJIE MAO

Software Development Engineer

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EDUCATION

University of Southern California	Los Angeles, USA
Master of Science in Computer Science GPA - 3.81/4.00	Sep 2023 – May 2025
University of Michigan - Ann Arbor	Ann Arbor, USA
Bachelor of Science in Computer Science (Dual Degree) GPA - 3.78/4.00	Sep 2021 – Apr 2023
Shanghai Jiao Tong University	Shanghai, China
Bachelor of Science in Electrical and Computer Engineering (Dual Degree) GPA - 3.62/4.00	Sep 2019 – Aug 2023

TECHNICAL SKILLS

Languages: Python, C/C++, SystemVerilog, JavaScript, TypeScript, Java, Kotlin, Bash Scripting, HTML, CSS
Frameworks & Development Tools: Linux, CUDA, NeuronxDistributed, NodeJS, React, Angular, Bootstrap,Pytorch
Databases: SQLite, Oracle, MongoDB

WORK EXPERIENCE

Amazon Web Services	San Jose, USA
Software Dev Engineer Intern - Machine Learning Chip Accelerator LLM, torchXLA	May 2024 – Aug 2024
- Implemented multi-modal model inference on AWS ML chips (Trn1/Inf2) under NeuronxDistributed framework. - Enabled input and output channel level tensor parallel convolutional layer on neuron device. - Designed preprocessing of the embedding mask on CPU to avoid tensor materialization and IR graph truncation.	
Shanghai Jiao Tong University	Shanghai, China
Engineering Drawing OCR Systems Research Assistant Python, PaddleOCR, PyTorch	Dec 2022 – Nov 2023
- Augmented data according to test results, including the generation of annotations with complex background. - Added triplet vector embedding techniques to PaddleOCR workflow, improving recognition accuracy to over 80%.	

PROJECTS

Responsive Web & Android App Dev Typescript, NodeJS, Angular, MongoDB, Kotlin, REST API	Jan 2024 – May 2024
- Developed a real-time stock search website with responsive design using Bootstrap and Angular framework. - Deployed NodeJS backend server and MongoDB Atlas to persistently store user information. - Enabled gesture control features on migrated android stock search application.	
Reliable Transport Protocol & BBR Congestion Control C/C++, Socket Programming, Quagga, BBR	Sep 2023 – Dec 2023
- Built a reliable transport protocol on top of UDP with cumulative acknowledgment and sliding window to resolve issues of packet loss, delay, corruption, duplication, and reordering. - Configured OSPF, iBGP and eBGP within an Autonomous System using Quagga routing suite. - Added Bottleneck Bandwidth and Round-trip propagation time(BBR) congestion control protocol to simplified TCP.	
R10K-style Out-of-Order Superscalar Processor SystemVerilog	Jan 2023 - Apr 2023
- Oversaw the intricate design of critical components of an R10k-style out-of-order superscalar processor, including the Reorder Buffer (ROB), Reservation Stations (RS), and Physical Register File (PRF) to increase instruction level parallelism. - Implemented a multiport non-blocking instruction cache and an instruction fetcher coordinating with branch predictor. - Engineered a 2-way set-associative data cache featuring a victim cache under LRU policy, to minimize cache misses.	
Fakebook Database Oracle SQL, MongoDB, Java, JavaScript	Jan 2023 – Apr 2023
- Designed an ER Diagram and built a schema of tables from a Facebook-like application database. - Constructed SQL queries using Java and JDBC to extract useful information from fakebook database. - Migrated tables in fakebook database to a MongoDB collection of users and finished queries on the collection.	
Optimization of the Neural Network Convolutional Forward Pass CUDA C/C++	Sep 2022 – Dec 2022
- Adopted shared memory tiling technique to reduce global memory traffic of the forward path of convolutional layers. - Reduced convolution kernel to a highly efficient matrix multiplication kernel with input unfolding and replicating. - Exploited multiple fine-tuned kernels tailored for various convolutional layers within deep learning models.	
Operating System Emulation C/C++, Multithreading, Network File System	Sep 2022 – Dec 2022
- Customized a thread library with mutex support to manage uniprocessor scheduling. - Implemented virtual memory management pager, optimizing memory allocation and resource utilization. - Orchestrated emulation of a multi-threaded, multi-client, and secure network file system server, minimizing disk I/Os.	